# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

| | | | |
|---|---|---|---|
| **Course Code:** | DSA03 | **Course Title:** | Natural Language Processing |
| **CO Assessed:** | CO5 – Articulation | **Assessment:** | Assessment Tool 1 – Constraint-Based Problem Solving |
| **Weightage:** | 35% of CIA | | |
| **CO5 Statement:** | Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5) | | |
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| **Section / Batch:** | AIDS/2024 | **Date:** | |

**Code of Conduct**

*I M.KASICF JALAL(192424214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the StudenT : kasif jalal M

**Instructions**

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

| Section / Topic | Focus | Question No. |
|---|---|---|
| Discourse, Reference Resolution & Text Coherence | Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph. | Q1 |
| Dialog and Conversational Agents, Discourse Planning & Surface Realizations | Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output. | Q2 |
| Machine Translation, Discourse Structure, Predicate Logic & Interpretation | Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases. | Q3 |

## Annexure A – Constraint-Based Problem Solving

1. **Given the following paragraph:**
   "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

   **Tasks:** a) Identify all referring expressions (mentions) and possible antecedents.

   b) Apply the following **constraints** to resolve the references:

   - Gender and number agreement
   - Recency (prefer recent antecedents)
   - Semantic compatibility (e.g., animacy and action suitability)
   - Coherence (maintain consistent entity chains)

   c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

   d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

   e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. **Conversation History:** User: "I have an important exam tomorrow but I'm not able to concentrate."

   **Required Dialog Act:** Advise + Encourage (Give advice and motivate the student).

   **Constraints:**

   1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
   2. Use **discourse relations** (Cause-Effect or Elaboration).
   3. Include at least **two** of these keywords: focus, break, confident.
   4. Response length: 2–3 sentences only.
   5. Logical consistency (no contradictory statements).
   6. Politeness and positive tone.

   **Tasks:** a) Identify the relevant discourse planning steps needed before generation.

   b) Generate **three** different possible responses that satisfy **all** the above constraints.

   c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

   d) Explain how violating any two constraints would affect the quality of the dialog.

3. **Source Sentence:** "The bank by the river flooded after the storm, but it was saved by quick action."

   **Note:** "bank" is ambiguous (financial institution or riverbank).

   **Constraints:**

   1. Correct **Word Sense Disambiguation** using context.

2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

**Tasks:** a) Resolve the ambiguous word "bank" and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

**1. Given the following paragraph:**

**"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."**

**Tasks: a) Identify all referring expressions (mentions) and possible antecedents.**

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**c) Draw a constraint graph or table showing variables, domains, and which constraints eliminate possible links.**

**d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.**

**e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.**

**Given Paragraph :**

“John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home.”

**a) Referring expressions and possible antecedents**

| Referring Expression | Possible Antecedents | Final Antecedent |
|---|---|---|
| **He** | John, Mary | **John** |
| **She** | John, Mary | **Mary** |
| **it** | ball, park, dog | **ball** |
| **him** | John, Mary, dog | **John** |
| **they** | John, Mary, dog | **John, Mary, dog** |
| **all** | John, Mary, dog | **John, Mary, dog** |

The assessment specifically asks us to consider **gender/number agreement, recency, semantic compatibility, and coherence** while resolving these references.

**b) Applying the constraints**

**1. Gender and Number Agreement**

**“He”**

- John → male → compatible
- Mary → female → incompatible

Therefore:

**He → John**

**“She”**

- John → male → incompatible
- Mary → female → compatible

Therefore:

**She → Mary**

**“it”**

- “it” is singular and generally refers to a non-human object.
- “ball” is singular and non-human.
- Therefore:

**it → ball**

**“him”**

- “him” must refer to a male person/animal.
- John is male.
- Mary is female.
- Dog is grammatically compatible as male/animal only if male is established, which it is not.

Thus, the strongest resolution is:

**him → John**

**“they”**

- “they” is plural.
- John + Mary + dog form the group.

Therefore:

**they → John + Mary + dog**

**c) Constraint Graph / Table**

A simple constraint representation can be written as follows:

```
        HE
      /    \
  John      Mary
   ✓         ✗
   \
```

```
   ↓
 JOHN


     SHE
    /    \
 John    Mary
  ✗       ✓
          ↓
        MARY


     IT
    /    \
  ball   park
   ✓      ✗
   ↓
 BALL


     HIM
   /   |   \
 John  Mary  dog
  ✓     ✗     ?
  ↓
 JOHN
```

```
            THEY
         /    |     \
     John    Mary     dog
         \     |     /
           GROUP
```

**Constraint table**

| **Mention** | **Candidate** | **Gender/Number** | **Semantic Compatibility** | **Coherence** | **Result** |
|---|---|---|---|---|---|
| He | John | ✓ | ✓ | ✓ | **John** |
| He | Mary | ✗ | ✓ | ✗ | Eliminated |
| She | John | ✗ | ✓ | ✗ | Eliminated |
| She | Mary | ✓ | ✓ | ✓ | **Mary** |
| it | ball | ✓ | ✓ | ✓ | **ball** |
| it | park | ✓ | ✗ | ✗ | Eliminated |
| him | John | ✓ | ✓ | ✓ | **John** |
| him | Mary | ✗ | ✓ | ✗ | Eliminated |
| they | John + Mary + dog | ✓ | ✓ | ✓ | **Group** |

**d) Final Coreference Chains**

**Coreference Chain 1 – John**

**John → He → him → they**

**Coreference Chain 2 – Mary**

**Mary → She → they**

**Coreference Chain 3 – Ball**

**ball → it**

**Coreference Chain 4 – Dog**

**dog → they**

So the final entity structure is:

JOHN ———→ He ———→ him ———→ they

MARY ———→ She ——————————→ they

BALL ———→ it

DOG ——————————————————→ they

**Rewritten Paragraph**

**John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.**

This version removes ambiguity and makes every reference explicit.

**e) Priority of Constraints**

The best priority order is:

**1. Gender and Number Agreement**

This should have high priority because it immediately eliminates impossible candidates.

Example:

- "He" cannot refer to Mary.
- "She" cannot refer to John.

**2. Semantic Compatibility**

The meaning of the sentence should be considered.

For example:

"She wanted to play with it."

A **ball** is semantically appropriate as something someone can play with, while a park is less appropriate as the direct object of "play with it" in this context.

**3. Coherence**

The selected references should create a consistent story.

The entities should remain consistent throughout the paragraph.

**4. Recency**

Among candidates that satisfy the stronger constraints, the most recently mentioned suitable entity can be preferred.

The assessment explicitly identifies **recency as a constraint**, with the instruction to prefer recent antecedents.

#### If Recency Is Relaxed

If recency is relaxed, the system will depend more strongly on:

- gender and number agreement,
- semantic compatibility,
- discourse coherence.

This can actually be useful when the closest noun is not the correct antecedent.

For example, **"He"** should refer to John even though other entities may appear nearby, because gender and discourse context strongly support John.

#### Conclusion for Q1

Constraint-based coreference resolution is effective because multiple linguistic constraints work together to eliminate incorrect references and maintain a consistent entity chain.

## 2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

**Required Dialog Act: Advise + Encourage (Give advice and motivate the student).**

**Constraints:**

- **Maintain entity coherence (reuse "exam", "concentrate", "you").**
- **Use discourse relations (Cause-Effect or Elaboration).**
- **Include at least two of these keywords: focus, break, confident.**
- **Response length: 2–3 sentences only.**
- **Logical consistency (no contradictory statements).**
- **Politeness and positive tone.**

**Tasks: a) Identify the relevant discourse planning steps needed before generation.**

**b) Generate three different possible responses that satisfy all the above constraints.**

**c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.**

## d) Explain how violating any two constraints would affect the quality of the dialog.

## Given Conversation

**User:**

"I have an important exam tomorrow but I'm not able to concentrate."

**Required Dialog Act**

**Advise + Encourage**

The response must:

- maintain the entities **exam, concentrate, you**,
- use a Cause-Effect or Elaboration relation,
- contain at least two of **focus, break, confident**,
- contain only **2–3 sentences**,
- be logically consistent,
- have a polite and positive tone.

**a) Discourse Planning Steps**

Before generating the response, the following steps should be performed.

**Step 1: Identify the user's intention**

The user is expressing difficulty concentrating before an exam.

Main intention:

**Need for advice and encouragement.**

**Step 2: Identify important entities**

The important entities are:

- **exam**
- **you**
- **concentrate**

These should be retained in the generated response.

**Step 3: Select the dialog acts**

Two dialog acts are required:

1. **Advise** – provide a practical suggestion.
2. **Encourage** – motivate the student.

**Step 4: Select discourse relation**

A **Cause-Effect** relation can be used:

Take a short break → improve concentration.

Or an **Elaboration** relation:

Give advice → explain how it helps.

**Step 5: Check keywords**

At least two of:

- focus
- break
- confident

must be included.

**Step 6: Surface realization**

Generate a natural, polite and positive 2–3 sentence response.

**b) Three Possible Responses**

**Response 1**

**"Take a short break and then return to your exam preparation with a clear focus. A few focused study sessions can help you concentrate better and feel more confident."**

Keywords:

- break ✓
- focus ✓
- confident ✓

**Response 2**

**"Take a short break, relax, and then study one topic at a time so you can focus better on your exam. You can concentrate more effectively when you avoid trying to study everything at once."**

Keywords:

- break ✓
- focus ✓

**Response 3**

**"Try studying in short sessions with small breaks so you can maintain your focus for the exam. Stay confident because consistent preparation can help you concentrate better."**

Keywords:

- break ✓
- focus ✓
- confident ✓

**c) Evaluation and Best Response**

| Constraint | Response 1 | Response 2 | Response 3 |
|---|---|---|---|
| Advise | ✓ | ✓ | ✓ |
| Encourage | ✓ | ✓ | ✓ |
| Exam maintained | ✓ | ✓ | ✓ |
| Concentrate maintained | ✓ | ✓ | ✓ |
| Focus | ✓ | ✓ | ✓ |
| Break | ✓ | ✓ | ✓ |
| Confident | ✓ | ✗ | ✓ |
| 2–3 sentences | ✓ | ✓ | ✓ |
| Positive tone | ✓ | ✓ | ✓ |
| Logical consistency | ✓ | ✓ | ✓ |

**Best Response: Response 1**

**Response 1 is the best response** because it satisfies all the constraints and combines advice with encouragement naturally.

It has:

- **Advice:** take a short break.
- **Cause-effect relation:** focused sessions can improve concentration.
- **Encouragement:** feeling more confident.
- **Required keywords:** break, focus, confident.
- **Entity coherence:** exam, concentrate, you.
- **Correct length:** 2 sentences.
- **Positive tone:** encouraging and polite.

Therefore, **Response 1 provides the strongest overall constraint satisfaction.**

**d) Effect of Violating Two Constraints**

**Violation 1: Length Constraint**

Suppose the response contains 5–6 sentences instead of 2–3.

**Effect:**

- The answer becomes unnecessarily lengthy.
- It may not satisfy the assessment requirement.
- The dialog becomes less efficient.
- Important information may be hidden among unnecessary details.

**Violation 2: Keyword Constraint**

Suppose the response does not contain at least two of:

**focus, break, confident**

**Effect:**

- The response fails an explicit constraint.
- Even if the advice is meaningful, it does not completely satisfy the required specification.
- The generated response receives a lower constraint-satisfaction score.

**Conclusion for Q2**

Constraint-based dialogue generation ensures that the response is not only grammatically correct but also satisfies **dialog act, discourse, content, length, keyword, coherence, and**

## 3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

**Note: "bank" is ambiguous (financial institution or riverbank).**

**Constraints:**

- **Correct Word Sense Disambiguation using context.**
- **Create a predicate logic representation (e.g., using predicates like flood(x), location(x,y)).**
- **Maintain discourse structure (Contrast relation between the two clauses).**
- **Preserve all important entities and relations (no information loss).**
- **Ensure text coherence in the generated output.**

**Tasks: a) Resolve the ambiguous word "bank" and justify using constraints.**

**b) Write a predicate logic representation of the sentence.**

**c) Generate a good target language sentence (in any Indian language or simple English paraphrase) that satisfies all constraints.**

**d) Draw a simple discourse tree (RST-style) showing the rhetorical relation.**

**e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence**

**Source Sentence**

"The bank by the river flooded after the storm, but it was saved by quick action."

The assessment identifies **"bank" as ambiguous**, because it can mean either a **financial institution** or the **riverbank**.

**a) Word Sense Disambiguation**

The word **bank** has two major possible meanings:

**Sense 1: Financial Institution**

Example:

I deposited money in the bank.

**Sense 2: Riverbank**

Example:

The riverbank flooded after heavy rain.

In our sentence:

**"The bank by the river flooded after the storm..."**

The phrase **"by the river"** provides strong contextual evidence.

A financial institution does not normally "flood" in the intended sense, while a **riverbank can flood**.

Therefore:

**bank = riverbank**

**Constraint-Based WSD**

| **Constraint** | **Financial Bank** | **Riverbank** |
|---|---|---|
| "by the river" | ✗ | ✓ |
| Can flood | Less compatible | ✓ |
| Storm context | Less compatible | ✓ |
| Semantic compatibility | ✗ | ✓ |
| Overall coherence | ✗ | ✓ |

Therefore:

**Bank → Riverbank**

The context eliminates the financial sense.

**b) Predicate Logic Representation**

We can represent the sentence using predicates such as:

- bank(x) – x is a bank/riverbank
- river(y) – y is a river
- by(x,y) – x is beside y
- storm(s) – s is a storm
- after(f,s) – flooding occurred after the storm

- flood(x) – x flooded
- saved(x,a) – x was saved by action a
- quick(a) – action a was quick

**Predicate representation**

bank(x)

river(r)

by(x,r)

storm(s)

flood(x)

after(flood(x), s)

quick(a)

saved(x,a)

A more compact logical representation is:

∃x ∃r ∃s ∃a [

RiverBank(x) ∧

River(r) ∧

LocatedBy(x,r) ∧

Storm(s) ∧

Flood(x) ∧

After(Flood(x),s) ∧

Quick(a) ∧

Saved(x,a)

]

This represents the important entities and relationships without losing the meaning of the original sentence.

**c) Target Sentence / Paraphrase**

A good simple English paraphrase is:

**“The riverbank near the river flooded after the storm, but quick action saved it.”**

An even clearer version is:

**“The riverbank flooded after the storm, but it was saved because people acted quickly.”**

This preserves:

- riverbank meaning,
- storm,
- flooding,
- quick action,
- contrast between the two clauses.

The assessment requires the generated sentence to preserve the important entities and relations and maintain coherence.

**d) RST-Style Discourse Tree**

The sentence contains two clauses connected by **“but.”**

**Clause 1**

**The riverbank flooded after the storm.**

**Clause 2**

**It was saved by quick action.**

The rhetorical relationship is:

**CONTRAST**

```
          CONTRAST

            |

 |              |

Clause 1       Clause 2

 |              |
```

```
 The riverbank            It was saved

 flooded after           by quick action

 the storm.
```

Or:

```
             [CONTRAST]
               /    \
              /      \
             ↓        ↓
    [Riverbank flooded]  [Riverbank saved]
          |                  |
     after the storm       by quick action
```

The word **“but”** explicitly signals the contrast relation, as required in the question.

**e) Advantages of Constraint-Based Approach over Pure Statistical MT**

**1. Better Word Sense Disambiguation**

A constraint-based system can explicitly use:

“by the river”

to select **riverbank** instead of financial institution.

A purely statistical system may choose the wrong sense if the context is insufficient or unusual.

**2. Preserves Semantic Meaning**

Constraint-based processing explicitly checks whether the selected interpretation is semantically compatible.

For example:

river + bank + flood

strongly supports **riverbank**.

**3. Maintains Discourse Structure**

The system recognizes:

Clause 1 + BUT + Clause 2

as a **Contrast** relationship.

Therefore, the translated sentence can preserve the original rhetorical meaning.

**4. Prevents Information Loss**

Important information such as:

- bank/riverbank,
- river,
- storm,
- flooding,
- quick action

can be explicitly represented as constraints.

**5. Produces More Explainable Decisions**

A major advantage is that the system can explain **why** it selected a particular interpretation:

“Bank” is interpreted as riverbank because it occurs near “river” and is semantically compatible with “flooded.”

This is easier to interpret than simply receiving a statistical probability.

**6. Handles Ambiguity Explicitly**

The system can generate candidate interpretations:

bank → financial institution

bank → riverbank

and then eliminate candidates using constraints.

### Rubrics for Calculation

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Problem Understanding | 25% | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps | Partial understanding; misses key constraints | Fails to identify constraints correctly |
| Constraint Analysis | 25% | Analyzes constraints effectively and prioritizes them logically | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization | No meaningful analysis or prioritization |
| Solution Development | 25% | Develops optimal and feasible solutions satisfying all constraints | Develops workable solutions with minor limitations | Solutions partially satisfy constraints | Solutions do not meet constraints |
| Application of Concepts | 15% | Effectively applies appropriate concepts, methods, or tools | Applies concepts with minor errors | Limited application; requires guidance | Unable to apply relevant concepts |
| Justification | 10% | Provides strong justification for decisions with logical reasoning | Provides justification with some clarity | Weak or unclear justification | No justification provided |

| Q. No. / Criteria Level | Problem Understanding | Constraint Analysis | Solution Development | Application of Concepts | Justification | Sub. Total | Total % |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |

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|---|---|---|
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| Date: ____________________ | Date: ____________________ | Date: ____________________ |